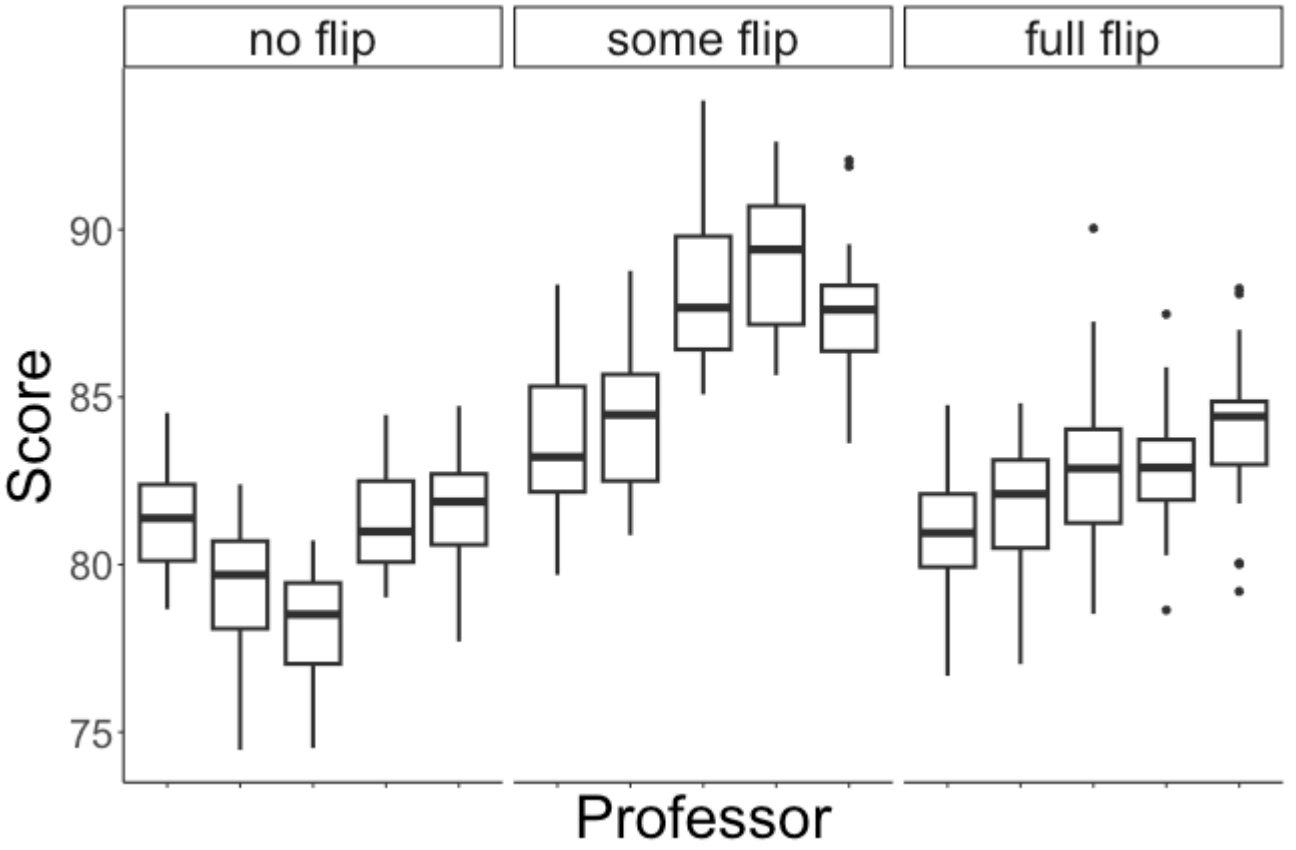
Beginning linear mixed effects models

Data: flipped classrooms?

Data set has 375 rows (one per student), and the following variables:

- `professor`: which professor taught the class (1 -- 15)
- `style`: which teaching style the professor used (no flip, some flip, fully flipped)
- `score`: the student's score on the final exam

Visualizing the data



Mixed effects model

Linear mixed effects model: Let $Score_{ij}$ be the score of student j in class i

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

Class activity

Work on the class activity (handout), then we will discuss as a class.

Class activity

> Why is a mixed effect model useful for this data?

- neighborhood is probably related to price, so want to include neighborhood
- 43 different neighborhoods → 42 new β terms (a lot!)

We want to avoid all those βs

- For inference, we don't care about comparing neighborhoods

Class activity

> What is the population model?

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

i = neighborhood

j = rental in neighborhood i

u_i = differences between neighborhoods

ε_{ij} = differences between rentals within a neighborhood

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

Class activity

> What is the population model?

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

- u_i is a *random intercept*
- We use subscripts i and j for $Price_{ij}$, $Satisfaction_{ij}$, and ε_{ij} because they are different for every observation in the data
- We only need a subscript i (neighborhood) for u_i, because there is one random intercept per neighborhood

Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$ $\quad$ $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

where $Price_{ij}$ is the price of rental j in neighborhood i

> What are the effect of interest, group effect, and individual effect?

Class activity

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

> What are the effect of interest, group effect, and individual effect?

- effect of interest: β_1 (slope for relationship between satisfaction and price)
- group effect: u_i (random effect for neighborhood)
- individual effect: ε_{ij} (variation between rentals in a neighborhood)

Fitting mixed effects models

```
library(lme4)
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

- `lme4` is the R package we will use to fit mixed effects models
- `lmer` is like the `lm` function, but for mixed effects
- `style` is the teaching style (fixed effects)
- `(1|professor)` indicates we have a random intercept (the `1` indicates the intercept) for `professor` (indicated by `|professor`)

Fitting mixed effects models

```
library(lme4)
```

```
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

```
Random effects:
 Groups    Name        Variance Std.Dev.
 professor (Intercept) 21.365   4.622
 Residual                4.252   2.062
Number of obs: 375, groups:  professor, 15
```

- $\widehat{\sigma}^2_\varepsilon = 4.25$
- $\widehat{\sigma}^2_u = 21.37$

Fitting mixed effects models

```
m1 <- lmer(score ~ style + (1|professor),
           data = teaching)
summary(m1)
```

```
Fixed effects:
                Estimate Std. Error t value
(Intercept)       77.657      2.075  37.419
stylesome flip    11.073      2.935   3.773
stylefull flip     2.805      2.935   0.956
```

- $\widehat{\beta}_0 = 77.66$
- $\widehat{\beta}_1 = 11.07$
- $\widehat{\beta}_2 = 2.81$

Interpretation

For class i, No flip group: average score on final is $\beta_0 + u_i$
average of $u_i = 0$ $(u_i \sim N(0, \sigma_u^2))$
expected final score over all classes in No flip group is β_0

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad u_i \overset{iid}{\sim} N(0, \sigma_u^2)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}_\varepsilon^2 = 4.25, \quad \widehat{\sigma}_u^2 = 21.37$

> How do I interpret $\widehat{\beta}_0 = 77.66$?

The expected test score for student in "no flip" classrooms is estimated to be 77.66 (averaging across all students & classes)

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \qquad \widehat{\beta}_1 = 11.07, \qquad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \qquad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_1 = 11.07$ and $\widehat{\beta}_2 = 2.81$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_1 = 11.07$ and $\widehat{\beta}_2 = 2.81$?

We expect that, on average, scores in some-flipped classes are 11.07 points higher than for no-flipped classes.

We expect that, on average, scores in fully-flipped classes are 2.81 points higher than for no-flipped classes.

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \qquad \widehat{\beta}_1 = 11.07, \qquad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \qquad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_u = 21.37$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \qquad \widehat{\beta}_1 = 11.07, \qquad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \qquad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_u = 21.37$?

The average score varies from professor to professor by about 4.62 ($= \sqrt{21.37}$) points

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_\varepsilon = 4.25$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

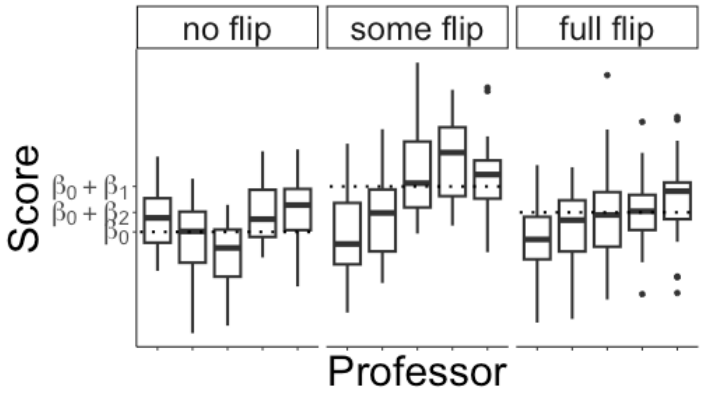
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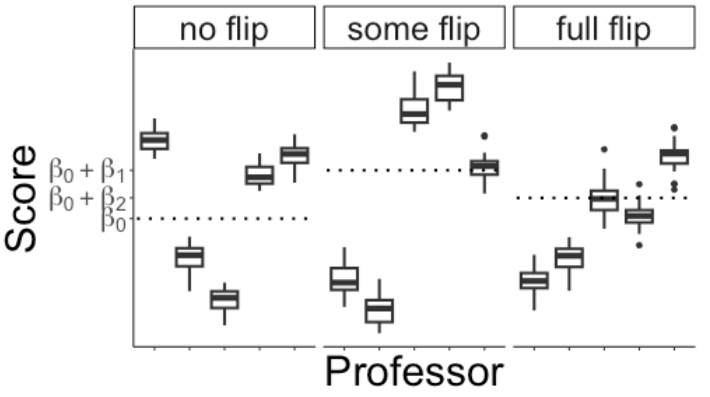
> How do I interpret $\widehat{\sigma}^2_\varepsilon = 4.25$?

Within a class, students' scores vary by about 2.06 ($= \sqrt{4.25}$) points

(we are making a constant variance assumption that σ_ε^2 is the same for all classes)

Intra-class correlation





σ^2_ε is large relative to σ^2_u

σ^2_ε is small relative to σ^2_u

- Observations within a group are *more correlated* when σ^2_ε is small relative to σ^2_u
- **Intra-class correlation:**

$$\rho_{group} = \frac{\sigma^2_u}{\sigma^2_u + \sigma^2_\varepsilon} = \frac{\text{between group variance}}{\text{total variance}}$$

Intra-class correlation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \quad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

$$\hat{\rho}_{group} = \frac{21.37}{21.37 + 4.25} = 0.83$$

So 83% of the variation in student's scores can be explained by differences in average scores from class to class (after accounting for teaching style). That's huge!

Class activity

Work on the class activity (handout).

Class activity

Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

Class activity

> Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

On average (across neighborhoods), we expect that the price of rental with 0 overall satisfaction is $27.28.

For a fixed neighborhood, an increase of 1 point in overall satisfaction is associated with an increase of $14.81 in rental price.

Class activity

> Calculate and interpret the estimated intra-class correlation.

Class activity

> Calculate and interpret the estimated intra-class correlation.

$$\hat{\rho}_{group} = \frac{1048}{1048 + 6762} = 0.134$$

About 13% of the variability in price can be explained by differences in the average price between neighborhoods (after accounting for overall satisfaction).